Scenerio Based AI

Interview Questions

5	43	∞
Sections	Scenarios	Value

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Full answers + explanations — Link in caption

WHAT'S INSIDE

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RAG (Retrieval-Augmented Generation)
9 scenarios

CrewAI
10 scenarios

SECTION 1 · PART 1 OF 2

GenAI Apps using Plain Python

1. **Retrieval Pipeline Failure**
   You built a RAG chatbot in plain Python (chunking + embeddings + FAISS). Users say answers are wrong even though the information exists in the PDF. How would you debug chunking strategy, embedding quality, retrieval scoring, and final prompt design?

2. **Non-Deterministic Model Behavior**
   Your LLM returns different outputs for the same input in production. What causes this behavior, and how would you make the system deterministic?

3. **Structured Output Integrity Breakdown**
   Your "JSON-only" response still includes extra text. How would you enforce strict JSON formatting and implement safe retry logic?

4. **Escalating Embedding Costs**
   Embedding expenses are rising rapidly. What caching or deduplication strategy would you design, and where would you store embeddings?

Full answers + explanations — Link in caption

SECTION 1 · PART 2 OF 2

GenAI Apps using Plain Python

5 **Custom Tool-Oriented Agent Design**

You need tool-calling without LangChain (calculator, web search, vector search). How would you design the tool schema, argument parsing, and re-invocation loop?

Full answers + explanations — Link in caption

SECTION 2 · PART 1 OF 3

LangChain

6 **Retrieval Quality Degradation**

You built a RAG app using LangChain's RetrievalQA. Users report irrelevant answers. How would you debug retriever configuration, chunking strategy, and prompt template?

7 **Incorrect Tool Selection**

Your LangChain agent using tools sometimes selects the wrong tool. How would you improve tool descriptions and better control agent reasoning?

8 **Memory Saturation Issue**

You are using ConversationBufferMemory. After long chats, responses degrade. Why does this happen and which memory strategy would you switch to?

9 **Structured Output Instability**

Your StructuredOutputParser frequently fails due to formatting errors. How would you make output parsing more reliable?

• **Full answers + explanations — Link in caption**

SECTION 2 · PART 2 OF 3

LangChain

10 **High Latency in Production**

You deployed a LangChain app and latency is high. How would you optimize chains, parallelize calls, or reduce token usage?

11 **Hybrid Retrieval Architecture**

You need hybrid search (BM25 + vector store) in LangChain. How would you design the retriever pipeline?

12 **Agent Infinite Loop Problem**

Your agent loops endlessly while using tools. How would you configure max iterations and stopping conditions?

13 **Streaming Response Integration**

You must add streaming responses in a LangChain app. Where would you enable streaming and how would you handle callbacks?

Full answers + explanations — Link in caption

SECTION 2 · PART 3 OF 3

LangChain

14 **Observability and Debugging**

You need to log intermediate reasoning steps for debugging. How would you use LangSmith or callbacks?

15 **Offline Enterprise Deployment**

You must deploy LangChain in an enterprise without internet. What components would you replace or self-host?

Full answers + explanations — Link in caption

SECTION 3 · PART 1 OF 2

LangGraph

16 **Missing Node Execution**

You built a multi-step workflow in LangGraph. Sometimes it skips a node. How would you debug state transitions, edges, and conditional checks?

17 **Cyclic Execution Loop**

Your LangGraph agent loops endlessly between two nodes. How would you design stopping conditions, max hops, and guardrails?

18 **Shared State Collision**

You are using shared state across nodes. A bug causes overwritten values. How would you structure state updates safely to avoid collisions?

19 **Unstable Conditional Routing**

You need routing based on LLM output. How would you implement deterministic branching and reduce ambiguity in decisions?

Full answers + explanations — Link in caption

SECTION 3 · PART 2 OF 2

LangGraph

20 **Node Timeout Resilience**

A node fails due to API timeout. How would you implement retries, backoff, and failure recovery inside the graph?

21 **Persistent Memory Requirement**

You must persist conversation state across sessions. How would you design checkpointing and state rehydration?

22 **Parallel Research Orchestration**

You want parallel execution of two research nodes before summarization. How would you structure the graph for fan-out and fan-in?

23 **Human Approval Gate**

You need human-in-the-loop approval before final output. How would you pause execution, capture feedback, and resume the graph?

24 **Performance Bottleneck Investigation**

Your graph latency is high. How would you profile node-level performance and identify bottlenecks?

• Full answers + explanations — Link in caption

SECTION 4 · PART 1 OF 3

RAG (Retrieval-Augmented Generation)

25 **Partial Answer Accuracy**

You built a RAG system, but answers are only partially correct. How would you isolate whether the issue is retrieval quality or generation behavior?

26 **Missing Context Retrieval**

Users complain the system misses relevant context. How would you redesign chunk size, overlap, and chunk boundaries for better recall?

27 **Redundant Retrieval Results**

Your retriever returns too many similar chunks. How would you improve diversity in results while keeping relevance high?

28 **Similarity Metric Decision**

You are choosing between cosine similarity and dot product. How would you decide which to use based on embeddings and indexing setup?

Full answers + explanations — Link in caption

SECTION 4 · PART 2 OF 3

RAG (Retrieval-Augmented Generation)

29 **Rising Embedding Spend**

Embedding costs are rising rapidly. What optimization strategies would you implement to reduce cost without hurting quality?

30 **Grounding Failure and Hallucinations**

Your RAG system hallucinates beyond retrieved context. How would you restrict generation to use only retrieved data?

31 **Multi-Document Attribution**

You must support multi-document comparison. How would you maintain reliable source attribution in responses?

32 **Context Window Limitations**

Long documents exceed context limits. How would you implement hierarchical retrieval or progressive summarization?

Full answers + explanations — Link in caption

RAG (Retrieval-Augmented Generation)

33

Explainability Requirement

You need explainability in RAG answers. How would you show which chunks influenced the response and why they were selected?

Full answers + explanations — Link in caption

SECTION 5 · PART 1 OF 3

CrewAI

34 **Inconsistent Multi-Agent Output**

You built a CrewAI system with Researcher and Writer agents. Output quality is inconsistent. How would you redesign roles, goals, and task descriptions to stabilize results?

35 **Redundant Agent Contributions**

Agents keep producing overlapping content. How would you improve task delegation, boundaries, and reduce duplication?

36 **Infinite Reasoning Loops**

Your crew enters repeated reasoning loops. How would you enforce stopping rules, max iterations, and convergence checks?

37 **Missing Context Handoff**

One agent depends on another's output, but context is missing. How would you design shared memory and handoff formats?

Full answers + explanations — Link in caption

SECTION 5 · PART 2 OF 3

CrewAI

38 **Review Gate Workflow**

You must add a Reviewer agent to validate output before final delivery. How would you design the workflow and acceptance criteria?

39 **Slow Sequential Execution**

Crew execution is slow due to sequential tasks. How would you parallelize independent tasks and merge outputs safely?

40 **Tool Integration Safety**

You need tool access (web search, calculator). How would you safely integrate tools into CrewAI agents with validation and constraints?

41 **Debugging and Observability**

You must log intermediate outputs for debugging. How would you implement observability across agents and tasks?

Full answers + explanations — Link in caption

SECTION 5 · PART 3 OF 3

CrewAI

42 **Hallucination Control**

An agent hallucinates facts. How would you introduce validation steps, fact-checking, and confidence gating?

43 **Production Deployment Readiness**

You must deploy CrewAI in production. How would you manage scaling, retries, failure handling, and error recovery?

Full answers + explanations — Link in caption

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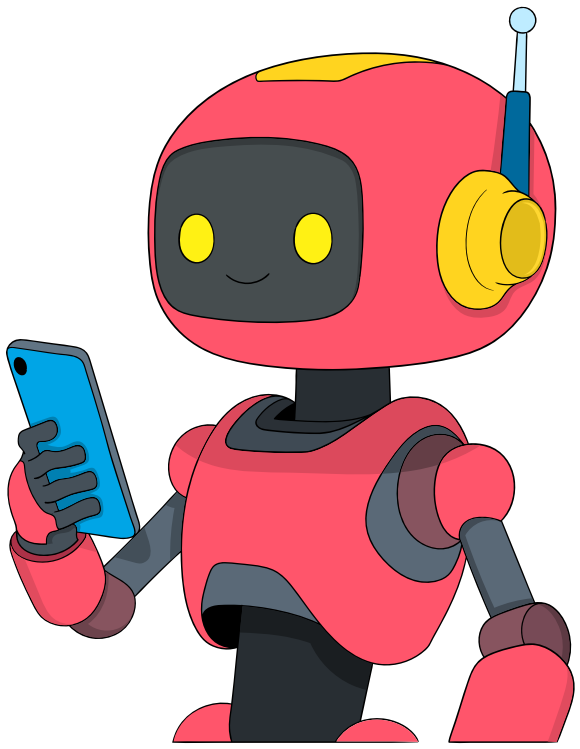

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